ABHILASH D G

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PROFESSIONAL SUMMARY

Neo4j Developer with 5 years of experience in Java, Spring Boot, Cypher, and graph data modeling for telecom network inventory. Skilled in Ciena Blue Planet Inventory (BPI), Metadata Modeler, Kafka-based ETL pipelines, data migration from SQL and PostgreSQL to Neo4j, Drools rules, inventory federation and reconciliation, REST APIs, and CI/CD.

SKILLS

Graph Database: Neo4j, Cypher Query Language, Graph Data Modeling, Graph Database Migration

Databases: Neo4j, PostgreSQL, MySQL, SQL

Blue Planet Inventory (BPI): Blue Planet Inventory (BPI), Metadata Modeler, Federation and Reconciliation, Discovery Resource Adapter (Discovery RA), Network Inventory

Programming: Java, Spring Boot, Python

Rule Engine: Drools (DRL)

Data Integration and Messaging: Kafka, ETL Pipelines, Data Migration, REST API, JSON

Orchestration: MDSO Resource Adapters (RA), Service Templates (ST), TOSCA, Jinja2, BPMN, Camunda

DevOps and Tools: Docker, Kubernetes, CI/CD, Git, GitLab, Jira, Nagios, Linux, Windows

Practices: SDLC, Agile, Scrum

PROFESSIONAL EXPERIENCE

Software Engineer — Apr 2025 - Present
Tech Mahindra, Bangalore — *Project: Telefonica-MDOI Germany – Netsite Federation (BPI / Neo4j)*

- Designed and developed graph data models and the **Netsite Federation** ingestion platform on **Neo4j** using **Java/Spring Boot** and **Cypher**, federating telecom network data (locations, cells, virtual containers, networks) into the **Blue Planet Inventory (BPI)** graph to support complex network relationships and analytics.
- Built Java-based event-driven ingestion and data connectors using **Kafka** and the Discovery Resource Adapter (Discovery RA); enriched discovered data with Netsite-specific properties and propagated it to the **Plan and Live views** alongside Netsite inventory.
- Implemented **Drools (DRL)** rule-engine logic for node deduplication and data consistency in the Neo4j graph, ensuring clean, conflict-free entities across multiple data sources.
- Led data migration of key network entities (locations, virtual containers, cells, networks) from SQL to **PostgreSQL** with snapshot-based comparison and validation, then loaded them into the **Neo4j** graph-based inventory using **Cypher**, ensuring data accuracy and consistency.
- Applied inventory federation and reconciliation techniques to integrate federated and discovered data; analyzed reconciliation reports and debugged discrepancies to keep graph data consistent across sources.
- Designed and developed **CoCheApp**, an in-house application to compare and analyze inventory data across Netsite, ENDB, and MDOI, with automation scripts, comparison logic, and an interactive dashboard that improved cell and network data quality from **80% to 99.3%**; added automated daily email reporting for customers.

- Ingested data from ENDB and validated it against offline inventory datasets, updating Neo4j node properties via **Cypher** for devices, cards, cells, and virtual containers to support accurate NMS migration and inventory synchronization.

System Engineer — Jun 2024 - Mar 2025
Tata Consultancy Services, Bangalore — *Blue Planet Inventory / MDSO*

- Modeled network devices and services in **Blue Planet Inventory** using Metadata Modeler, creating L3VPN and VRF service representations to ensure accurate service-to-resource relationships in the inventory.
- Designed and implemented end-to-end **Camunda** workflows for IPVPN, orchestrating service provisioning and inventory updates.
- Designed use cases for IP Connect, VLAN configuration, and IP VPN with the Provisioning Resource Adapter; built a GNS3 router topology (SSH-enabled) for testing.
- Explored modeling of NTN communication architecture (5G, gNB) in the inventory.

Software Engineer — Aug 2021 - Jun 2024
Prodapt Solutions, Bangalore — *Blue Planet MDSO / Orchestration*

- Led a cross-functional team at Frontier Communications to design, develop, and deliver **Blue Planet MDSO** solutions, building Resource Adapters and Service Templates from scratch using Python, JSON, Jinja2, and TOSCA.
- Upgraded the Blue Planet orchestration environment at Charter Communications from **19.10 to 21.10**, including Resource Adapters and Service Templates.
- Integrated BPMN/Camunda workflows with MDSO via TOSCA interfaces for Service Order Orchestration (SOO) and automated provisioning.
- Containerized applications with Docker and Kubernetes; improved DevOps practices in Linux environments.
- Contributed to the SLATEST project: device provisioning, service activation, and L3SAT / Y.1564 testing.

EDUCATION

Bachelor of Engineering, BTL Institute of Technology (2017-2021)
Information Science Engineering, CGPA: 8.02/10.00

RECOGNITIONS

- **Spotlight Award:** Completed the Blue Planet 21.10 upgrade with a good understanding of the upgrade procedure. Sep 9th, 2022
- **Spotlight Award:** Efficiently handled all aspects of the Frontier HBE project, from analysis to UAT, and received customer appreciation. June 6th, 2023

CERTIFICATIONS

- **Blueplanet Sales Orchestration** — Ciena Blueplanet
- **Docker CI/CD** — Udemy